

Evaluating Lightweight Domain Adaptation for Host-based Intrusion Detection

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Abstract—Intrusion detection systems (IDS) powered by machine learning can experience significant performance degradation when deployed in environments that differ from the data on which they were originally trained. While domain adaptation (DA) has emerged in intrusion detection as a promising solution to this problem, most prior research assumes centralized network-based architectures and relies on computationally expensive deep learning methods. Few works pursue the possibility of DA deployed to endpoints in host-based IDS (HIDS) which operate under stricter resource constraints and face unique forms of domain shift due to device heterogeneity, variable workloads, and differences in user behavior.

This paper investigates how effectively a minimal-complexity DA layer based on the Correlation Alignment (CORAL) approach can reduce cross-domain performance degradation in HIDS. We develop a test environment and CORAL calculation methodology with the goal of incorporation in endpoint HIDS, assessing several classification models trained on the CIC-IDS2017 dataset as a source domain. To evaluate cross-domain performance, the CSE-CIC-IDS2018 dataset as well as captured university computer lab traffic serve as target domain datasets. Our results show that lightweight CORAL statistical alignment was unable to effectively improve cross-domain detection performance in the evaluated HIDS setting, instead causing macro recall to collapse across all classifiers and target domains. These findings suggest that simple global covariance alignment is insufficient for preserving minority attack-class structure under significant domain shift among heterogeneous traffic, despite its low computational overhead and endpoint-deployment appeal.

Index Terms—intrusion detection system, domain adaptation, CORAL, host IDS, machine learning, anomaly detection

I. INTRODUCTION

The growing scale and sophistication of cyberattacks has increased demand for intrusion detection systems (IDS) capable of identifying malicious activity across diverse and constantly changing environments [1] [2]. IDS machine learning (ML) models frequently experience performance degradation when deployed outside the environments in which they were originally trained due to cross-domain distribution variations [3] [4] [2]. This shift between training and deployment conditions reduces model accuracy and limits the practical usefulness of otherwise effective detection systems [4] [2].

Domain adaptation (DA) has emerged as a promising approach for addressing this issue by reducing the distribution mismatch between source and target domains [3] [5]. Most existing DA approaches for IDS attempt to learn domain-invariant feature representations that reduce statistical discrepancies between source and target environments in order to improve detection performance of source-trained models

on target data. [6] [7] [5]. A major approach in the space is adversarial learning, in which two models compete by generating examples and learning to distinguish them from real data. Generative Adversarial Networks (GANs) proposed in [8] was particularly foundational in adversarial learning DA. The approach shaped other influential DA techniques such as Domain-Adversarial Neural Networks (DANN), which learn features that support prediction while obscuring domain origin, and Adversarial Discriminative Domain Adaptation (ADDA), which extended this direction by aligning target samples with the source feature space without requiring labeled target data [6] [7]. More broadly, these foundational adversarial DA architectures, along with other techniques such as feature-alignment methods, have influenced many recent approaches for learning transferable representations across shifted domains [5]. While these methods can achieve strong cross-domain performance, deep neural network-based approaches are frequently computationally expensive, difficult to deploy, and poorly suited to resource-constrained endpoint systems [1] [5].

Most existing DA research for IDS also assumes a centralized network intrusion detection system (NIDS) architecture in which traffic is aggregated and processed at a central location. NIDS monitor network traffic at the network level, whereas HIDS monitor host-level activity such as incoming and outgoing packets on individual machines [1]. In such settings, higher computational overhead is generally acceptable because the underlying infrastructure is expected to support more demanding workloads. Comparatively fewer works have explored domain adaptation in host-based intrusion detection systems (HIDS), particularly lightweight HIDS architectures intended for deployment directly on endpoint machines.

Although less common in the literature, DA at the host level is valuable for several reasons. Endpoint hosts often exhibit highly personalized behavior due to differences in installed applications, workload patterns, hardware characteristics, user activity, and network usage. As a result, the distribution shift experienced by a specific host may differ substantially from the broader network-level patterns learned by centralized NIDS DA approaches. A host-based DA system therefore offers the potential for more localized adaptation and more precise detection performance increases for a given machine. Additionally, performing adaptation directly on the endpoint can improve privacy by avoiding the need to aggregate host traffic or behavioral data to a centralized server. A decentralized architecture

may also improve scalability by distributing computational work across multiple endpoints rather than concentrating it within a single network bottleneck.

To address these challenges, this study explores the feasibility of light-weight DA in intrusion detection with a particular focus on compatibility with HIDS endpoint restrictions. The proposed system includes a minimal-complexity processing layer based on statistical alignment method Correlation Alignment (CORAL), then uses several classifiers for prediction on target domain traffic. The study evaluates CORAL across two target domains. The first consists of the CSE-CIC-IDS2018 benchmark dataset, which differs from the source-domain CIC-IDS2017 dataset used for model training. The second target domain consists of organically captured traffic from a university computer lab. This evaluation design allows the study to examine both transfer performance across public benchmark datasets and generalization to real-world network conditions.

The primary contributions of this work are as follows:

- The development of a lightweight CORAL-based DA layer suitable for integration into a deployed resource-constrained HIDS application.
- An evaluation of whether a simple statistical alignment method can effectively reduce cross-domain performance degradation.
- A cross-environment evaluation framework spanning both public benchmark datasets and organically captured real-world network traffic in order to assess performance under realistic deployment conditions.

II. RELATED WORKS

This section reviews previous research on DA for IDS, focusing on prominent approaches in the literature and highlighting lightweight statistical feature alignment methods such as CORAL. We also examine resource-aware host-based IDS architectures relevant to deployable endpoint intrusion detection, and connect these discussions to the specific gap that our work addresses.

A. DA Approaches in IDS

DA in IDS research overwhelmingly consists of deep learning techniques in which novel architectures are proposed to improve cross domain generalization. Using such approaches, many studies aim to address the problem of labeled target-domain data scarcity, notably in the context of Internet of Things (IoT) traffic. One such work is [9], which proposed MRADDA-TC, a multi-source adversarial DA framework that combines several feature extraction and classification techniques to improve intrusion detection across heterogeneous IoT environments with limited training samples. Similarly, [10] introduced E-ADDA, an enhanced adversarial DA architecture that improves cross-dataset IDS generalization using stronger feature encoding, regularization, and uncertainty estimation methods. The work of [11] also addressed the issue of limited target data availability by proposing FIADDA,

a few-shot adversarial DA framework designed to improve performance in IoT environments.

Other works emphasize adversarial representation learning as the central mechanism for constructing domain-invariant feature spaces. [12] investigated a DANN-based IDS framework that models network flows as packet-byte images to enable adversarial feature alignment between heterogeneous domains. [13] proposed DA-GAN, which combines generative adversarial networks with adversarial DA to mitigate labeled data scarcity while improving generalization. [14] introduced DI-NIDS, a domain-invariant IDS architecture that incorporates one-class anomaly detection into adversarial DA for improved resilience against unseen attacks and distributions. [15] similarly focused on adversarial DA for preparing deep-learning-based network IDS with limited target data. Collectively, these works illustrate the broader IDS research direction of combining deep neural feature extraction with adversarial optimization to learn generalized domain-invariant representations.

More recent research has moved beyond adversarial approaches by combining DA with other techniques such as self-supervised learning, graph neural networks, and concept-drift handling to improve generalizability. For example, [16] proposed ReCDA, which combines self-supervised feature learning and adaptive classifier tuning to maintain performance under evolving traffic patterns. Similarly, [17] developed a framework using graph convolutional networks and feature distribution alignment, while a later work by the same researchers in [18] introduced graph attention and causal embedding methods for IoT domains. Likewise, [19] proposed a self-supervised and domain-adaptive framework for healthcare IoT traffic using contrastive learning and adversarial adaptation. Additionally, the work in [20] proposes a hybrid approach named OLIDA-IDS that combines adaptation with continual "online" learning to address domain drift in dynamic environments.

Overall, IDS DA research remains dominated by computationally intensive deep learning architectures, and lightweight adaptation mechanisms suitable for resource-constrained systems remain comparatively underexplored.

B. CORAL and Statistical Feature Alignment

A less explored approach to DA in IDS is explicit statistical feature alignment for reducing domain shift in network traffic. CORAL is a particularly lightweight approach which differs from adversarial DA by not requiring complex neural architectures, adversarial optimization, or retraining procedures. Introduced in [21] and later expanded on in [22] for deep learning applications, CORAL reduces domain shift by aligning the covariance statistics of source and target feature distributions through lightweight second-order statistical alignment, making it well suited for resource-constrained deployment scenarios.

The work in [23] proposed a lightweight feature-driven domain adaptation framework for Internet of Medical Things intrusion detection that emphasizes reduced computational overhead while performing adaptation across heterogeneous

IoT domains. The work in [24] similarly highlights the utility of feature adaptation across heterogeneous IoT network environments through transfer learning approaches that avoid computationally intensive adversarial training.

However, the application of CORAL-style lightweight statistical adaptation to deployable host-based IDS environments remains limited in current literature.

C. Host-Based and Lightweight IDS Architectures

Host-based intrusion detection systems (HIDS) operate under stricter resource constraints than centralized network IDS architectures, making computational efficiency an important design requirement.

While most DA-focused IDS research evaluates centralized network-based systems, the work in [25] explores a domain-adaptive HIDS framework designed for low-resource environments while emphasizing practical deployment under resource constraints.

III. METHODOLOGY

This section presents the design choices behind the study's conducted procedures. Details regarding the particular DA method that this study implements are discussed in III-A, descriptions of the datasets used and data processing procedures are explained in III-B and III-C, and a justification for the classification models used is included in III-D.

A. CORAL and HIDS Integration

To address the problem of cross-domain performance degradation, this work employs CORAL as the DA technique. As mentioned previously, CORAL is an unsupervised DA method which aligns source and target feature distributions through covariance transformation. The method minimizes domain shift by aligning the second-order statistics of the source feature space to those of the target feature space, and generating a transformed version of the target domain for source-trained classifiers to predict more precisely on.

CORAL was selected due to several characteristics relevant to endpoint-based intrusion detection. First, CORAL operates as a lightweight statistical transformation layer rather than a computationally intensive neural architecture, making it suitable for resource-constrained endpoint environments. Second, the method is model agnostic, allowing transformed feature representations to be used with multiple downstream classifiers without requiring modifications to classifier architecture or training methodology. Additionally, the datasets used in this work satisfy CORAL's prerequisite assumptions of shared feature and label spaces across domains, as both employed public datasets were developed by the same researchers with consistent traffic collection tools.

In the context of a deployed HIDS application which implements a DA feature, CORAL is well-suited for integration due to its relatively low computational overhead and minimal adaptation requirements. The DA mechanism can be readily incorporated into the typical capture-to-prediction pipeline of an IDS as an additional preprocessing stage executed

prior to classifier inference. A prospective design for a DA-aided IDS application would incorporate user-initiated traffic capture sessions to collect target environment network traffic, perform target covariance estimation, then load precomputed source-domain statistics to generate the cross-domain CORAL transform. When applied, predictions made by source-trained classifiers on the CORAL-transformed target data are expected to show improved performance as opposed to without. Figure 1 shows a prospective design for integrating DA into an IDS processing pipeline, as well as how such a system relates to the design of the parallel offline experimental workflow used for this study.

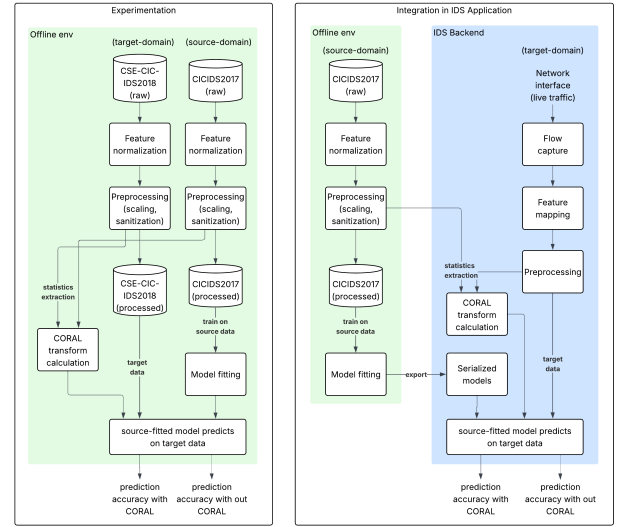


Fig. 1. CORAL-based DA IDS evaluation experiment design (left) compared to integration into an IDS pipeline (right).

B. Datasets

Three intrusion detection datasets were employed in this study to adequately assess CORAL's robustness in improving domain shift across diverse environments. All used datasets contained flow-based network traffic data of labeled benign or malicious flows.

For the experiment's source domain, the CIC-IDS2017 public dataset was used due to its prevalence in the literature and inclusion of diverse and modern attacks [26]. The experiment also consisted of two target domain datasets. First was the CSE-CIC-IDS2018 public dataset, an updated version of the CIC-IDS2017 which contains more complex and extensive samples [26]. Both datasets share highly similar feature schema and attack classes, making them suitable candidates for CORAL-based adaptation. The second target domain dataset used was comprised of computer laboratory-captured network traffic in order to evaluate real-world conditions. This lab data was captured in the CSU Northridge College of Engineering and Computer Science computer lab, and was collected with a custom IDS application deployed to a lab machine and running

its traffic-capture feature while the machine was undergoing a typical user workload. Because the lab machine endpoint captures occurred after firewall filtration, all naturally captured traffic was assumed to be benign.

C. Data Preparation

Data preprocessing procedures were designed to maintain consistency across datasets, preserve compatibility with CORAL's cross-domain shared schema requirements, and align with the design goal of possible incorporation into a deployable DA-aided HIDS system. In preparing the source and target datasets, similar preprocessing steps were applied with some important distinctions.

First of these differences was in the preparation of the CSECICIDS2018, which underwent significant size reduction due to its substantially larger amount of samples. The source dataset CICIDS2017 contained an original sample size of over 2.8 million flows, while the CSECICIDS2018 contained nearly 16 million. Meanwhile, the lab-captured data amounted to just 135,000 benign samples. To account for these discrepancies across dataset sizes, the CSECICIDS2018 target data was reduced in preprocessing to the approximate size and class distribution of the source CICIDS2017 to form a test scenario in which cross-domain data conditions are approximately similar. The data discrepancy of the lab-captured data, however, was preserved to simulate the likely conditions of a real-world scenario in which collected target data would be limited. For reduction of CSE-CIC-IDS2018, a hybrid capped/stratified sampling strategy was employed in which rare classes were preserved, medium classes were moderately reduced, and large classes were aggressively capped. Table I shows attack support across datasets after preprocessing.

Next, all datasets underwent data sanitization in which samples containing invalid or unusable values were removed, and data formats were standardized. This was followed by feature and label space alignment in which all datasets were verified to have equivalent schema as required by CORAL. Features and attack classes not universally shared across datasets were removed to form common feature and label spaces. The shared set of classes present across all datasets can be seen in Table I.

Following schema alignment, attack samples were extracted from the CSE-CIC-IDS2018 dataset and injected into the captured lab dataset in order to add malicious traffic to the otherwise benign-only samples collected from the lab. Attack samples from the first target domain were chosen for injection into the second to create consistency across evaluation scenarios. Total injected attack amounts and attack class frequencies were designed such that rare classes were preserved while larger ones were reduced. The benign/attack ratio of the resulting consolidated lab data was also engineered to be roughly 75/25, approximately that of the source domain proportion. This decision was made because although such a proportion of malicious traffic may be unrealistic to real-world HIDS target-data capture, it was important for attack patterns to be prevalent enough in the data such that a classifier could meaningfully learn patterns and achieve measurable

TABLE I
CLASS SUPPORT ACROSS DATASETS AFTER LABEL-SPACE ALIGNMENT.

Class	CIC-IDS2017	CSE-CIC-IDS2018	Lab Capture Data
Benign	TODO	TODO	TODO
Bot	TODO	TODO	TODO
DDoS	TODO	TODO	TODO
DoS GoldenEye	TODO	TODO	TODO
DoS Hulk	TODO	TODO	TODO
DoS	TODO	TODO	TODO
Slowhttptest			
DoS Slowloris	TODO	TODO	TODO
FTP-Patator	TODO	TODO	TODO
Infiltration	TODO	TODO	TODO
SSH-Patator	TODO	TODO	TODO
Web Attack - Brute Force	TODO	TODO	TODO
Web Attack - Sql Injection	TODO	TODO	TODO
Web Attack - XSS	TODO	TODO	TODO
Total	TODO	TODO	TODO

performance differences with and without CORAL. For this reason, attack samples were present in all datasets, and made to be at roughly equal benign/attack proportions. Benign/attack proportions for each dataset following preprocessing can be seen in Table II.

The next preprocessing step for all datasets was creating the train and test splits. An 80/20 train/test division was used for all datasets with a stratified split to preserve the original class balance across splits. This is because the CORAL transformation depends heavily on the distribution of the data from which covariance is extracted, so preserving class distribution across splits was crucial. CORAL statistics extraction was performed solely on the train split of all datasets in order to prevent leakage into the test split. Test splits were used to evaluate classifier performance with and without the CORAL transform. To avoid inflated results, the data used to learn the CORAL transform was kept separate from the data used for evaluation. Final benign/attack proportions, train/test split sizes, and data shape for all three datasets are shown in Table II.

TABLE II
DATASET CHARACTERISTICS AFTER PREPROCESSING.

Dataset	Shape (row, col)	Benign/Attack	Train/Test
CIC-IDS2017	(2430093, 78)	TODO%/TODO%	80%/20%
CSE-CIC-IDS2018	(2320849, 78)	73.99%/26.01%	80%/20%
Captured Lab Data	(182865, 71)	74.31%/25.69%	80%/20%

Note: Captured lab data is shown to have a reduced number of columns due to the discrepancy of flow features available to the differing capture technologies used. Lab data was collected using Nfstream, which supports fewer flow characteristics than CICFlowMeter which was used for collecting the other datasets.

After performing data splits, each dataset underwent scaling and label encoding in which a scaler and encoder were fitted and applied to the source domain, then subsequently reused and applied to both target domains. Finally, covariance

statistics were extracted from all datasets’ train splits to be used in calculating the CORAL transform, and processed data was exported. III shows a complete list enumerating the preprocessing steps performed on each dataset.

TABLE III
PREPROCESSING WORKFLOW FOR SOURCE AND TARGET DOMAIN DATASETS.

Step	Source: CIC-IDS2017	Target 1: CSE-CIC-IDS2018	Target 2: Captured Lab Dataset
1	Import raw CSV	Import raw CSV	Import raw JSON
2	–	Data reduction	–
3	data sanitization	data sanitization	data sanitization
4	–	–	inject attack samples (from Target 1)
5	feature-space alignment	feature-space alignment	feature-space alignment
6	Label-space alignment	Label-space alignment	Label-space alignment
7	–	extract attack samples (for Target 2)	–
8	Train/test split	Train/test split	Train/test split
9	Scaling (fit and apply)	Scaling (apply source fitted scaler)	Scaling (apply source fitted scaler)
10	Label encoding (fit and apply)	Label encoding (apply source fitted encoder)	Label encoding (apply source fitted encoder)
11	Extract CORAL stats (train split)	Extract CORAL stats (train split)	Extract CORAL stats (train split)
12	Export processed data	Export processed data	Export processed data

Note: Dashes indicate steps not applied.

D. Models and Training

All evaluated classifiers were trained exclusively on the CIC-IDS2017 source-domain training split in order to preserve the experimental objective of source-to-target domain evaluation under domain shift conditions. Three classification models were selected for experimentation: Random Forest (RF), Support Vector Machine (SVM), and Multilayer Perceptron (MLP). These models were chosen because they provide a representative spread across major ML learning paradigms, are prevalent in IDS research, and fit the goal of low computation for endpoint systems. Specifically, RF serves as a strong benchmark for tree-based models and is shown to have strong performance across ML IDS literature. SVM provides a baseline for kernel-based learning models, and MLP represents a neural-network approach.

IV. EXPERIMENTAL SETUP

This section describes how the studies experiments were conducted to effectively evaluate CORAL’s impact on cross domain performance. The hardware/software environment and technologies used are discussed first in IV-A, followed by a description of model hyperparameter configurations in IV-B. The specifics of the CORAL implementation is described in IV-C, and finally a summary of the study’s evaluation system is discussed in IV-D.

A. Implementation Environment

Data processing, model training, and experimentation were conducted in a Python 3.13.7 Jupyter Notebook 7.5.5 environment using the scikit-learn 1.8.0 library for implementing model training and evaluation [27]. The project was structured into the following three notebooks to organize procedures: source preprocessing, target preprocessing, and training/evaluation, and nondeterministic procedures were implemented with a fixed random seed of 42 to ensure consistency and reproducibility. The packages pandas 3.0.2, NumPy 2.4.4, and joblib 1.5.3 were used for data management, processing, and model serialization.

The custom IDS used to capture lab traffic for the second target dataset consisted of a python-based traffic collection pipeline, notably using Nfstream for data collection and pandas and NumPy for run-time feature mapping and data scaling [28]. While capturing traffic, pertinent actions executed on the host machine included: web authentication, application updates, dataset downloads, and git commands.

B. Model Configuration

The three employed models, RF, SVM, and MLP were all used in experimentation and evaluated on both target domains after being fitted to the CIC-IDS2017 source domain. Models received hyperparameter configurations according to a prior study conducted by the authors.

In that study, several models were trained on CIC-IDS2017 using an equal-effort validation procedure for configuration selection, and tested hyperparameter values were chosen according to relevant studies [29] [30]. IV shows the hyperparameter grids used for all models. Configurations were evaluated based on model recall on the prior study’s held-out validation split of CIC-IDS2017. The best performing configurations from that study were reused for hyperparameter configurations in this work’s employed models.

TABLE IV
HYPERPARAMETER VALUES

Model	Hyperparameter	Values Tested	Best Config.
RF	n_estimators	{100, 300}	100
	max_depth	{None, 20}	None
	min_samples_split	{2, 10}	2
	min_samples_leaf	{1, 5}	1
SVM	kernel	{linear, rbf}	linear
	C	{0.5, 1, 1.5}	1.5
MLP	hidden_layer_sizes	{(64,), (128,)}	64
	alpha	{0.0001, 0.001}	0.001

C. CORAL Configuration

CORAL was applied as a feature-space transformation to adapt the held-out test-split of the target domains before prediction by the source-trained classifier. CORAL statistics were estimated from the training splits of the source and target datasets to avoid leakage, and the resulting statistics were then used to calculate the transformation.

Source and target CORAL statistics were computed by extracting the feature-wise mean vector and covariance matrix of each domain. Covariance estimation was performed using the Ledoit-Wolf shrinkage estimator of sklearn to improve numerical conditioning relative to the empirical covariance matrix.

The adapted target-domain feature representation was calculated according to standard CORAL procedures [21]. For a target feature matrix X_t , the CORAL-transformed representation was computed by using the target-domain mean μ_t to center target samples, applying a covariance alignment matrix comprised of C_t and C_s (target and source covariance matrices, respectively), and shifting the samples into the source-domain mean μ_s . Matrix square-root and inverse-square-root operations were computed by symmetric eigendecomposition. Equation 1 shows the complete calculation for the CORAL-transformed target feature space X_t^{CORAL} .

$$X_t^{\text{CORAL}} = (X_t - \mu_t)C_t^{-1/2}C_s^{1/2} + \mu_s, \quad (1)$$

D. Evaluation Procedure and Metrics

Experimentation consisted of three evaluation conditions conducted on both target domains. First, all source-fitted models were evaluated on the source domain's held-out test split as a baseline reference. Next, models were evaluated on the target domain test split without the CORAL transformation applied to establish the initial source-to-target domain shift, then with the transformation applied to measure CORAL's ability to improve cross domain performance.

The performance of each model was measured by standard ML metrics including: accuracy and weighted and macro versions of precision, recall, and F1 score.

V. RESULTS & ANALYSIS

This section presents the experimental results obtained from evaluating all three classifiers across each experimental condition. Due to the significant class imbalance present in all datasets, our analysis emphasizes the metric macro recall in evaluating classifier performance, as recognizing limited-support attack classes is a strong sign of cross domain robustness and is critical to intrusion detection.

On the source-domain benchmark evaluation, all models achieved high overall performance. The RF and MLP models produced the strongest source-domain test results, with accuracies near or above 0.99 and substantially higher macro recall than the SVM. This confirms that the models were not underfitting the source data and that the source-domain classification task was learned effectively. However, source-domain performance did not transfer reliably to the target domains, demonstrating the presence of substantial domain shift.

Without CORAL, all three models exhibited degraded performance on both target datasets. Accuracy remained between approximately 0.69 and 0.75, and macro recall was much lower, ranging from approximately 0.13 to 0.22 across the no-CORAL target experiments. These results indicates that

although the models often correctly identified the majority benign class, their ability to recognize attack classes was limited. Among the no-CORAL experiments, MLP provided the strongest overall target-domain macro recall, particularly on the lab data target domain, suggesting slightly better robustness to domain shift than SVM and RF.

Contrary to expectations, applying CORAL did not improve cross-domain intrusion detection performance. In the CSE-CIC-IDS2018 target experiments, CORAL caused all three classifiers to collapse to a macro recall of approximately 0.0769. This value corresponds to successful recall for only one of the thirteen classes, indicating that models' prediction on the CORAL transformed data became a virtually benign-only classification. Similar behavior was observed on the lab data target domain. Although some CORAL experiments produced target accuracies near 0.74, this value closely matches the Benign class proportion in the target datasets, once again indicating that models overwhelmingly predicted benign for all samples. Therefore, the apparent accuracy preservation after CORAL does not represent improved detection capability; it instead reflects a majority-class prediction pattern and minority-class recall collapse. Table V presents these results in detail, and Fig. 2 shows the negative impact of CORAL on macro recall for both target domains.

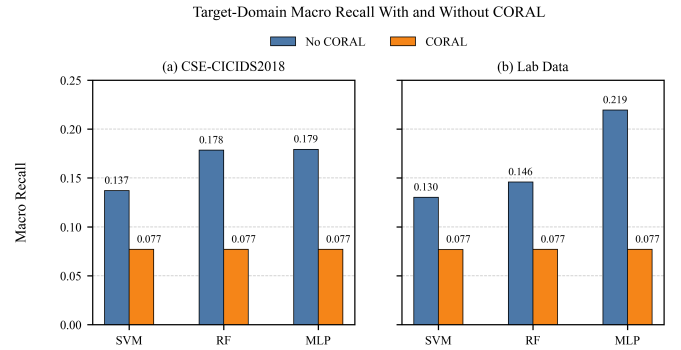


Fig. 2. Target-domain macro recall with and without CORAL on both target domains. CORAL reduced macro recall for all classifiers to approximately 0.077, indicating near-complete attack-class recall degradation.

The per-class reports further support this interpretation. After CORAL alignment, most attack categories fell to zero precision, zero recall, and zero F1-score. This suggests that the global covariance alignment performed by CORAL may have shifted target samples into regions of the feature space associated primarily with benign traffic, while failing to preserve class-discriminative structure for minority attacks. In contrast, the no-CORAL condition, although weak overall, retained partial detection ability for certain attack classes such as DoS GoldenEye, DoS slowloris, and Web Attack XSS. Fig. 3 presents the per class recall values produced by each classifier on both target domains, and shows the attack recall collapse after applying CORAL.

Overall, the results show that CORAL was ineffective for reducing cross-domain performance degradation in this

TABLE V
CLASSIFIER PERFORMANCE ACROSS TARGET DOMAINS WITH AND WITHOUT CORAL ADAPTATION.

Target Domain	Classifier	Source		Target w/o CORAL		Target w/ CORAL		Δ Macro Recall
		Accuracy	Macro Recall	Accuracy	Macro Recall	Accuracy	Macro Recall	
CSE-CIC-IDS2018	RF	0.9986	0.8453	0.7479	0.1785	0.7399	0.0769	-0.1016
CSE-CIC-IDS2018	SVM	0.9713	0.4576	0.7248	0.1371	0.7399	0.0769	-0.0602
CSE-CIC-IDS2018	MLP	0.9957	0.7218	0.7414	0.1794	0.7399	0.0769	-0.1025
Captured Lab Data	RF	0.9986	0.8484	0.7522	0.1458	0.7431	0.0769	-0.0689
Captured Lab Data	SVM	0.9723	0.4601	0.6881	0.1302	0.07429	0.0769	-0.0533
Captured Lab Data	MLP	0.9938	0.7154	0.7352	0.2195	0.7431	0.0769	-0.1426

Note: Δ Macro Recall indicates the difference between CORAL-adapted target macro recall and no-CORAL target macro recall, representing the impact of CORAL on improving cross domain recall degradation.



Fig. 3. Per-class target-domain recall for source-trained classifiers evaluated with and without CORAL. The no-CORAL condition retains partial recall for selected attack classes, while the CORAL condition collapses recall almost entirely to the Benign class.

experimental configuration. The best target-domain performance was achieved without CORAL, particularly by the MLP model. These findings suggest that unsupervised global feature alignment alone may be insufficient for multi-class intrusion detection under severe class imbalance and heterogeneous network traffic distributions.

VI. CONCLUSION

This paper investigated whether lightweight DA based on CORAL can reduce cross-domain performance degradation in HIDS. Unlike many prior DA IDS approaches which rely on computationally expensive deep learning techniques suited for NIDS architectures, this study evaluates a minimal statistical alignment layer intended to be compatible with resource-constrained endpoint deployment.

Three classifiers trained on source domain CIC-IDS2017 were evaluated in two target-domain settings (the CSE-CIC-IDS2018 dataset and a captured university computer lab traffic dataset augmented with attack samples) with and without CORAL to assess its impact on intrusion detection classification.

Reported results comparing benchmark source evaluation with no-CORAL target results indicate that considerable domain shift existed across both source-target pairs, however, CORAL was shown to be ineffective in reducing performance degradation. Contrary to initial expectations, CORAL reduced

macro recall for all classifiers and target domains to approximately 0.0769, indicating that prediction behavior collapsed almost entirely to the majority benign class. With CORAL applied, classifiers became largely benign-only predictors, causing inflated accuracy values with attack-class recall almost completely eliminated. These findings suggest that global unsupervised covariance alignment may be an insufficient solution for multi-class HIDS adaptation under severe class imbalance and heterogeneous traffic distributions, particularly when minority attack-class structure must be preserved.

The main contribution of this work is therefore a negative but practically important result: lightweight CORAL-style statistical alignment, although attractive for endpoint deployment due to its low computational complexity, did not provide reliable DA benefits in intrusion detection. This result highlights the need to evaluate lightweight adaptation methods not only by overall accuracy, but also by macro recall and per-class attack detection performance. In intrusion detection, preserving minority attack-class recall is essential, and an adaptation method that improves or maintains majority-class accuracy while degrading attack recognition is not suitable for deployment.

Several limitations of this study should be noted. The captured lab dataset contained naturally benign traffic and required injected attack samples from a public target dataset to support measurable attack classification. This design enabled controlled evaluation but may not fully represent real-world malicious activity in an endpoint lab environment. In addition, CORAL was applied as a global feature transformation across all classes, which may have distorted minority-class decision boundaries. The study also evaluated a limited set of classifiers and did not explore class-conditional alignment, online adaptation, or hybrid adaptation strategies.

Future work on CORAL for IDS should explore semi-supervised approaches using labeled target-domain samples for feedback in tuning parameters of the CORAL transformation to generate improved configurations. Another valuable direction would be to inject variable amounts of source domain samples into the target to assess the proportion of domain shift at which CORAL becomes effective. Additional experiments should also investigate other lightweight adaptation methods that better preserve class-discriminative structure to address the issue of minority class recall collapse. Furthermore, other

directions include evaluating more realistic host-captured attack traffic, broader endpoint environments, and analyzing runtime overhead of such a DA feature when incorporated within a deployed IDS application. Overall, the findings show that while lightweight DA remains a valuable direction for deployable HIDS, simple global CORAL alignment alone is not sufficient for robust cross-domain intrusion detection in the evaluated setting.

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